Skills Summary

Riemann Sums, Accumulation, and the Fundamental Theorem

Use this reference while completing your worksheet or when studying.

One chain. A Riemann sum *estimates* accumulated change from finitely many samples; its limit *is* the definite integral; the Fundamental Theorem then evaluates it exactly — and read backwards, differentiates it. **FTC 2:** $\int_a^b f(x)\,dx = F(b) - F(a)$. **FTC 1:** $\frac{d}{dx}\int_a^x f(t)\,dt = f(x)$.

1. Riemann sums as estimates

With n equal subintervals, $\Delta x = \dfrac{b-a}{n}$.

- **Left** sum: skip the last sample. **Right** sum: skip the first. Each uses n values.
- **Trapezoid:** $\dfrac{\Delta x}{2}\left[y_0 + 2y_1 + \cdots + 2y_{n-1} + y_n\right]$ — interior values count twice.
- If f is **increasing**, left underestimates and right overestimates; if decreasing, the reverse.

Example: A rate is sampled every 2 units as $3, 7, 10, 12$. Find the left Riemann sum over the three subintervals.
Step 1. A left sum uses the value at the start of each subinterval, so the last reading is not used at all — identify which three values that leaves.
Step 2. The left values are 3, 7 and 10 with $\Delta x = 2$, so the sum is $2(3 + 7 + 10) = 2(20)$.

$$\text{left sum} = \mathbf{40}$$

Try it: A rate is sampled every 3 units as $2, 5, 6, 9$. Find the left Riemann sum over the three subintervals.

check: **39**

2. Antiderivatives: reversing the derivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1); \quad \int \cos x\,dx = \sin x + C; \quad \int \sin x\,dx = -\cos x + C.$$

Antidifferentiate term by term, and check by differentiating — that check is exact and costs one line.

Example: Find $\displaystyle\int (3x^2 - 4x)\,dx$.
Step 1. The integrand is a sum of powers, so reverse the power rule on each term separately rather than looking for a substitution.
Step 2. $3x^2$ raises to $\dfrac{3x^3}{3} = x^3$ and $-4x$ raises to $-\dfrac{4x^2}{2} = -2x^2$.

$$\boldsymbol{x^3 - 2x^2 + C}$$

Try it: Find $\displaystyle\int (8x^3 + 2)\, dx$.

check: $\boldsymbol{2x^4 + 2x + C}$

3. Evaluating definite integrals (FTC 2)

Find any antiderivative F, then compute $F(b) - F(a)$. The $+C$ cancels, so one antiderivative is enough.

Order matters: $F(a) - F(b)$ gives the negative of the answer. A negative result is not an error — the definite integral measures **signed** area, and regions below the axis count negatively.

Example: Evaluate $\displaystyle\int_1^4 (2x + 1)\, dx$.

Step 1. The integrand has an elementary antiderivative, so FTC 2 applies and the work is one antiderivative plus one subtraction.
Step 2. $F(x) = x^2 + x$, so $F(4) = 20$ and $F(1) = 2$, and the integral is $20 - 2$.

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Try it: Evaluate $\displaystyle\int_0^2 3x^2\, dx$.

check: **8**

4. Accumulation functions (FTC 1)

$F(x) = \int_a^x f(t)\, dt$ is a *function of its upper limit*: net accumulation from a to x, with $F(a) = 0$. By FTC 1, $F'(x) = f(x)$, so F increases where f is positive.

For a quantity with a known rate, $A(t) = A(0) + \int_0^t r(u)\, du$ — the integral gives the **change**, never the amount.

Example: $F(x) = \displaystyle\int_0^x 2t\, dt$. Find $F(5)$.

Step 1. The upper limit is the input, so evaluating F at 5 means integrating from 0 to 5 — the variable t is internal and never appears in the answer.
Step 2. An antiderivative of $2t$ is t^2, so $F(5) = 25 - 0$. (By FTC 1, $F'(x) = 2x$, which is positive for $x > 0$, so F is increasing.)

$F(5) = \mathbf{25}$

Try it: $G(x) = \displaystyle\int_0^x 3t^2\, dt$. Find $G(2)$.

check: **8**